Enhancing Contextual Compatibility of Textual Steganography Systems Based on Large Language Models

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This systematic literature review examines the transformative impact of Large Language Models (LLMs) on linguistic steganography. Through comprehensive analysis of 26 primary studies, the research demonstrates that LLM-based approaches significantly enhance imperceptibility, embedding capacity, and naturalness in cover text generation, addressing traditional limitations of low embedding capacity and cognitive imperceptibility. The findings reveal a paradigm shift towards context-aware steganographic systems that leverage domain-specific knowledge and communicative context to achieve both perceptual and statistical imperceptibility. The review establishes that understanding contextual compatibility and domain correlations is crucial for developing more sophisticated, robust, and secure covert communication systems, paving the way for future advancements in generative text steganography.

Additional Key Words and Phrases: Systematic Literature Review, Linguistic Steganography, Large Language Models, LLMs, Natural Language Processing, NLP, Black-box Steganography, Context Retrieval, Generative Text Steganography, Imperceptibility

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1 INTRODUCTION

Linguistic steganography hides secrets inside ordinary sentences-an exploit that looks trivial until one remembers how little redundancy natural language actually contains [1, 2]. A single awkward synonym, a statistically rare clause, or an out-of-place idiom is enough to alert an automated sentry. Classic tricks-swap a word here, bend the syntax there-carry so few bits and leave such distinctive fingerprints that modern steganalysis routinely catches them [3].

Large language models change the game. Their uncanny fluency lets them spin entire documents that read like human prose yet obey an adversarial agenda: every plausible continuation is also a potential codeword.

None of these victories is absolute. Push the embedding rate and the text begins to creak; optimize for statistical stealth and the throughput collapses-the so-called "Psic effect" [1]. Still, progress is slow. This survey dissects the advances, catalogs the open wounds, and maps the territory that remains to be claimed.

This systematic review fills these gaps by meticulously identifying and synthesizing recent primary literature that leverages LLMs for textual steganography. The importance of this review is underscored by the transformative impact of LLMs on secure communication [citation/reference needed], marking a paradigm shift toward context-aware, generative systems that prioritize imperceptibility, embedding capacity, and naturalness [citation/reference needed].

The rest of the paper is structured as follows. Section 2 lays the theoretical groundwork by covering steganography, LLM capabilities, and the unique challenges of generative linguistic systems, including the Perceptual-Statistical Imperceptibility Conflict (PSIC). Section 3 reviews prior surveys to contextualize our contribution. Section 4 outlines our systematic review methodology-research questions, search strategy, and inclusion criteria. Section 5 presents our findings across five research questions (RQ1–RQ5), examining publication trends, applications, evaluation metrics,

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knowledge integration, and technical trade-offs. Section 6 synthesizes these results, discussing practical implications and ethical concerns. Finally, Section 7 concludes with key insights and directions for future research.

2 BACKGROUND

Information security systems broadly encompass **encryption**, **privacy**, and **concealment**, the last of which-known as **steganography**-is the focus of this review. While encryption and privacy protect message content, they do not conceal the existence of communication, which may itself arouse suspicion. Steganography instead prioritizes **imperceptibility**: embedding information into ordinary carriers (e.g., images or text) so that hidden messages remain unnoticed.

Text is a particularly challenging carrier due to its low redundancy and strict semantic constraints. The classical "Prisoners' Problem" [4] illustrates the goal: two parties, Alice and Bob, must exchange hidden information without alerting a watchful adversary.

Textual steganography methods are typically divided into **format-based** approaches, which exploit layout or structural features, and **content-based** approaches, which modify linguistic form. Within the latter, early techniques such as **synonym substitution** embed bits by altering lexical choices, but suffer from low capacity and high detectability. More formally, **linguistic steganography** refers to concealing information in natural language by modifying or generating text while preserving fluency and meaning [5].

Traditional linguistic approaches offer limited embedding capacity and often leave statistical artifacts. Advances in deep learning and **Large Language Models (LLMs)** now enable generative methods that achieve higher text quality and more secure embedding. Evaluating such systems requires several dimensions of imperceptibility: **perceptual** (human naturalness), **statistical** (distributional similarity to natural text), and **cognitive** (semantic and contextual fidelity) [6].

A deeper theoretical perspective introduces **channel entropy**, which quantifies the information-carrying capacity of a given communication channel. Entropy sets the upper bound for embedding rates: higher entropy allows more hidden information without detection, while lower entropy restricts capacity. Achieving this bound securely requires **perfect samplers**, which can generate text indistinguishable from genuine distributional samples. These concepts underpin the design of provably secure steganographic systems.

However, LLMs [7] introduce new challenges. Their tendency toward **hallucinations** can create detectable artifacts, highlighting the **Psic Effect** (Perceptual-Statistical Imperceptibility Conflict) [1], where optimizing for perceptual fluency may undermine statistical security. Model access further shapes practical steganography: **black-box access** (e.g., commercial APIs or hosted open-weight models) offers significant advantages, delivering substantially better text quality through access to state-of-the-art models, faster generation speeds via optimized infrastructure, and minimal local resource requirements, enabling scalable deployment without the computational overhead of training or hosting large models locally. The primary trade-off is limited control and reduced transparency over internal sampling probabilities. In contrast, **white-box access** enables fine-grained control over parameters and sampling, supporting stronger security guarantees, but typically demands substantial computational resources, engineering effort, and higher latency, raising deployment barriers. This trade-off is central to evaluating the robustness and applicability of modern linguistic steganography.

2.1 Capabilities and Approximating Natural Communication

Large Language Models (LLMs) are autoregressive, generative systems based on the Transformer architecture [8] that approximate high-dimensional distributions over natural-language sequences [2][9]. Given a prefix, an LLM emits a

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probability vector over the vocabulary; the next token is sampled from this vector and appended to the prefix, and the process repeats until a stopping criterion is met. During pre-training, billions of parameters are tuned on large text corpora so that the model's predictive distribution converges to the empirical distribution of the data [10]. As a consequence, modern LLMs routinely produce text whose fluency, coherence and style are indistinguishable from human writing [11]. The learned latent representations capture stylistic and semantic regularities that generalize across domains, enabling applications requiring nuanced linguistic mimicry [12].

2.2 Role in Generative Linguistic Steganography

LLMs are considered **favorable for generative text steganography** due to their ability to generate high-quality text. Researchers propose using generative models as steganographic samplers to embed messages into realistic communication distributions, such as text. This approach marks a departure from prior steganographic work, motivated by the public availability of high-quality models and significant efficiency gains.

LLMs like **GPT-2** [9], **LLaMA** [13], and **Baichuan2** [14] are commonly used as basic generative models for steganography. Existing methods often utilize a language model and steganographic mapping, where secret messages are embedded by establishing a mapping between binary bits and the sampling probability of words within the training vocabulary. However, traditional "white-box" methods necessitate sharing the exact language model and training vocabulary, which limits fluency, logic, and diversity compared to natural texts generated by modern black-box LLM APIs or hosted open-weight models. **Black-box approaches provide substantial advantages:** they deliver significantly better text quality by leveraging state-of-the-art hosted models, achieve faster generation speeds through optimized cloud infrastructure, and require minimal local resource requirements without the computational overhead of running large models locally. In contrast, white-box methods require running large models locally, increasing latency and resource consumption. These methods further inevitably alter the sampling probability distribution, thereby posing security risks [15].

New approaches, such as **LLM-Stega** [15], explore **black-box generative text steganography using the user interfaces (UIs) of LLMs**. This circumvents the requirement to access internal sampling distributions. The method constructs a keyword set and employs an encrypted steganographic mapping for embedding. It proposes an optimization mechanism based on reject sampling for accurate extraction and rich semantics [15].

Another framework, **Co-Stega**, leverages LLMs to address the challenge of low capacity in social media. It expands the text space for hiding messages through context retrieval and **increases the generated text's entropy via specific prompts** to enhance embedding capacity. This approach also aims to maintain text quality, fluency, and relevance [16].

The concept of **zero-shot linguistic steganography** with LLMs utilizes in-context learning, where samples of covertext are used as context to generate more intelligible stegotext using a question-answer (QA) paradigm [17]. LLMs are also employed in approaches like **ALiSa**, which directly conceals token-level secret messages in seemingly natural steganographic text generated by off-the-shelf BERT [18] models equipped with Gibbs sampling [19].

The increasing popularity of deep generative models has made it feasible for provably secure steganography to be applied in real-world scenarios, as they fulfill requirements for perfect samplers and explicit data distributions (see Section 2) [2, 20, 21].

LLM-based steganographic methods are typically evaluated on two primary axes: imperceptibility (perceptual, statistical, and cognitive measures) and embedding capacity (e.g., bits per token or bits per word). Imperceptibility evaluations may include automatic metrics (PPL, Distinct-n, MAUVE, KL/JSD) as well as human judgements; embedding capacity is usually reported as bits/token or overall embedding rate.

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We now turn to the principal challenges these models face, including the trade-off between imperceptibility and capacity, robustness to tokenization, and practical deployment constraints.

2.3 Challenges and Limitations in Steganography with LLMs

2.3.1 Perceptual vs. Statistical Imperceptibility (Psic Effect). The **Psic Effect** [1] represents a fundamental trade-off in steganographic systems. it's the inverse relationship between **text quality** and **resistance to statistical steganalysis** in generative steganography. Two components govern it:

- **Perceptual imperceptibility**: fluency/naturalness of a single sentence, gauged by human ratings or perplexity (PPL).
- **Statistical imperceptibility**: divergence between the distribution of stego and human text as measured by an automated steganalyzer.

Human social-media prose is casual and high-variance; it does not hug the optimal language-model peak. A generator that over-optimizes for quality produces text whose likelihood concentrates on that peak, yielding a detectable statistical spike against the broad, noisy human baseline.[1]

Experiments show that the most fluent stego sentences are the first ones caught by detectors, yet counter-intuitively pushing the embedding rate higher can make the text statistically safer because the added noise widens its distribution toward the authentic human scatter, a trade-off modern systems like VAE-Stega manage by learning to keep sentences smooth while staying inside the real variance envelope.[1]

2.3.2 Limited Embedding Capacity. Even with advanced generative capabilities, LLMs cannot overcome natural language's fundamental low-redundancy constraint, which limits embedding capacity. This issue is acute in common applications such as social media and dialogue systems, where communication is brief or predictable. In low-entropy scenarios (e.g., replying to "Happy birthday!"), the model has few natural-sounding alternatives. Consequently, steganographic techniques like rejection sampling face higher failure rates, and channel capacity for hiding data diminishes significantly [2, 16].

2.3.3 Poor Semantic Control and Contextual Drift. Early generative methods produced fluent but **semantically arbitrary** text, violating **cognitive imperceptibility** [6]. By conditioning only on preceding tokens, these models generated replies that drifted from the original logic-a mismatch that triggers immediate suspicion on social media (e.g., replying "Today is beautiful" to a technical post). Many white-box, sampling-based techniques face a deeper problem: they must continue generating until an arbitrarily large payload is fully embedded. The model may exhaust its topic and terminate prematurely, or the user might provide a broad topic to sustain output, risking unnaturally verbose text that is highly detectable outside long-form media like blog posts. This raises a critical question often unaddressed: given a large payload like an image, how do these methods handle segmentation and embedding across messages without violating contextual norms?

Generating long stego texts introduces further technical and security hurdles. Extended generation strains coherence and contextual consistency [22, 23], and minor early deviations compound over time to degrade decoding accuracy [24, 25]. This issue is magnified in low-entropy channels, where embedding even small messages requires impractically long cover text [2]. The resulting verbosity is not merely inefficient; on platforms like social media where brevity is the norm, abnormally long posts signal anomalous activity and increase detection risk [16]. Consequently, effective

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information load remains limited, with performance often diminishing when embedded messages exceed just a few tokens [25].

2.3.4 LLM-Specific Obstacles. Deploying steganography with LLMs introduces distinct challenges:

- **Computational Burden:** 3–5× higher time and resource costs versus prior neural methods
- **Black-Box Access:** Hosted APIs (whether proprietary or open-weight models) limit visibility into internal sampling probabilities, blocking white-box steganographic mappings. However, they provide significant advantages: access to state-of-the-art models that deliver substantially better text quality, faster generation speeds through optimized infrastructure, and minimal local resource requirements without the computational overhead of running large models locally
- **Hallucinations:** Factually incorrect or nonsensical output can corrupt the covert bitstream or create detectable patterns
- **Escalating Detection:** As LLM capabilities advance, so do machine learning-based **steganalysis** tools that distinguish synthetic from human text
- **Data Fragility:** Lossy compression or incomplete transmission of stegotext causes irreversible bitstream corruption

2.3.5 Tokenization Mismatch. Modern Transformer models using **subword tokenization** (e.g., BPE) suffer from **segmentation ambiguity**: a sender's token sequence (" any", "thing") may detokenize to "anything" but be **retokenized differently** by the receiver as a single token, " anything". This breaks the **autoregressive chain**, corrupting all downstream probability distributions and causing extraction failure. The problem is acute in **scriptio continua** languages like Chinese, which lack explicit word boundaries.

Analogy: Alice encodes a secret using two small bricks to spell "BLUE." Bob receives one large "BLUE" brick. Since their protocol depends on exact brick counts, Bob's misalignment renders the rest of the message unreadable.

Methods that rely on modifying the sampling probability distribution to embed secret messages inherently introduce security risks because they alter the original distribution, making the steganographic text statistically distinguishable from normal text [1, 2, 15, 20]. While advancements in deep neural networks have improved text fluency and embedding capacity, older models or certain embedding strategies can still produce texts that lack naturalness, logical coherence, or diversity compared to human-written content. Models like NMT-Stega and Hi-Stega aim to maintain semantic and contextual consistency by leveraging source texts or social media contexts, yet this remains a complex challenge [6, 26].

Channel entropy requirements and variability also pose a considerable challenge. Traditional universal steganographic schemes often demand consistent channel entropy, which is rarely maintained in real-world natural language communication. Moments of low or zero entropy can cause protocols to fail or require extraordinarily long steganographic texts. The Psic Effect highlights this dilemma in balancing quality and detectability.

Additional limitations include:

- **Data Integrity and Reversibility**: Some methods cannot perfectly recover the original cover text after message extraction [27, 28].
- **Ethical Concerns**: Pre-trained LLMs may introduce biases, discrimination, or inappropriate content [17, 29].
- **Provable Security**: Many NLP steganography works lack rigorous security analyses and fail to meet formal cryptographic definitions [2].

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3 RELATED REVIEWS

Majeed et al. (2021) [30] surveyed pre-LLM text steganography techniques, predating the current transformer era. Setiadi et al. (2025) [31] recognizes that LLMs have "revitalized" linguistic steganography, examining recent methods (2021-2025) using GPT-2 [32], GPT-3 [33], LLaMA2 [34], and Baichuan2 [35]. However, their review remains a critical examination rather than a systematic survey, leaving several key papers unaddressed. Crucially, the field has evolved from "statistical vector embedding" (Word2Vec, GloVe) to "language-model vector embedding" that exploits BERT-scale transformers and higher-dimensional semantic spaces.

This creates a methodological gap: no systematic review comprehensively maps how large-scale transformers have redefined text steganography. Modern advances extend beyond naive generation to sophisticated Controllable Text Generation (CTG) frameworks [36]. These employ Variational Autoencoders (VAEs) to model latent features and Diffusion Models to inject randomness, mitigating spurious associations between secrets and control conditions. Classical surveys emphasized synonym replacement, spacing manipulation, and Huffman coding [30]-techniques that predated LLMs. Earlier methods relied on context-free grammars (CFGs) or Markov chains, often producing syntactically correct but semantically incoherent cover texts. Contemporary approaches leverage prompt learning and prefix tuning, enabling efficient model customization without costly full fine-tuning.

Defensive strategies must evolve accordingly. Traditional steganalysis, premised on hand-crafted statistical features, falters against generative steganography's high statistical concealment. Current research must confront "stegomalware"-attacks that conceal command-and-control communications within innocuous digital media.

4 RESEARCH METHOD

This study was undertaken as a systematic mapping review using the guidelines presented in Petersen et al. [37]. The goal of this review is to identify, categorize, and analyze existing literature published between 2018 and 2025 and use syntactic and semantics aspects to represent context handling in linguistic steganographic methods.

4.1 Planning

In this section, we define our research questions, the search strategy we use, and the inclusion and exclusion criteria considered to filter the results.

4.1.1 Research Questions. This systematic literature review is guided by six research questions, aiming to comprehensively map the landscape of steganographic techniques leveraging large language models (LLMs). The questions explore the current state of published literature, applications where these techniques are being explored, and the metrics and evaluation methods used to assess their performance, with a focus on capacity, security, and contextual compatibility. Furthermore, the review investigates how external knowledge sources are integrated to enhance capacity or contextual relevance, the limitations and trade-offs associated with current techniques, and potential future research directions considering emerging trends and identified gaps.

4.1.2 Search Strategies. The initial literature search employed a specific query string: '(steganography or watermark or "Information Hiding") and ("Large Language Model" or LLM or BERT or LAMA or GPT)'. This query was executed across several digital libraries, including ACM Digital Library, IEEE Digital Library, Science@Direct, Scopus, and Springer Link, to ensure broad coverage. To complement this automated search and identify additional relevant studies, a snowballing technique was also applied. This involved examining the reference lists of included studies. While

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snowballing primarily yielded older steganographic techniques not explicitly mentioning LLMs, these papers often utilized similar methodological approaches to contemporary LLM-based steganography, providing valuable contextual information.

4.1.3 Inclusion and Exclusion Criteria. To ensure the selection of high-quality and relevant studies, the following criteria were applied.

Inclusion Criteria Studies were included if they:

IC1: Provided full-text access.
IC2: Were published in English from 2018 onwards.
IC3: Appeared in peer-reviewed journals, conferences, or workshops.
IC4: Directly addressed steganography, watermarking, or information hiding techniques involving or significantly impacted by LLMs, BERT, LAMA, or GPT architectures.
IC5: Represented empirical studies, surveys, reviews, or theoretical contributions.

Exclusion Criteria Studies were excluded if they:

EC1: Were duplicates (retaining the most complete or recent version).
EC2: Were incomplete, abstract-only, or irrelevant to steganography with LLMs.
EC3: Were non-English publications.
EC4: Came from non-peer-reviewed sources (e.g., preprints, dissertations, theses, books, book chapters), unless extended from peer-reviewed conference papers.

4.2 Conducting the Search

The initial automated search across the selected digital libraries yielded a total of 1043 candidate papers. The distribution by source was: ACM Digital Library (346), IEEE Digital Library (61), Science@Direct (209), Scopus (151), and Springer Link (276). Duplicated papers were automatically eliminated using Parsifal tool [1]. After removing all duplicates, 1,573 papers remained. Following this the papers underwent a multi-stage filtering process based on their titles, abstracts, and full texts, guided by the predefined inclusion and exclusion criteria. After title and abstract filtering, 58 papers remained. Of these, 26 were accepted with readily available PDFs, while 6 were pending PDF acquisition at the time of analysis.

4.3 Data Extraction and Classification

A Data Extraction Form (DEF) was developed to systematically collect data from each primary study to address our research questions. The form is designed in a table format consisting of the following types of information:

- Bibliometric Information: paper title, type (Steganography or Watermarking), author(s), publication year, and publication venue.
- Model Details: input and output formats, key characteristics, approach classification (three-term categorical), specific LLM used (if applicable), embedding process description, and code availability.
- Datasets: all datasets employed, including their sizes.
- Context Awareness: whether the method is "Explicit," "Implicit," or "No," the context keyword (e.g., "Social Media," "Formal Document"), how context is represented (e.g., "Text," "Pretext," "Graph," "Vector"), and how it is utilized in the method.

[1] https://parsif.al

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- Evaluation Details: evaluation metrics, steganalysis models used, and the best numerical results for each reported metric.
- Strengths and Limitations: main strengths and weaknesses of the approach or model.

Following data extraction, studies were classified based on predefined categories derived from the research questions to identify trends, patterns, and gaps in the literature. The results are summarized using tables, figures **??**), and descriptive statistics. Each research question is addressed individually with interpretation of findings and identification of future research directions.

5 RESULTS

This section presents the synthesized findings from our systematic literature review of 26 primary studies on LLM-based steganography. The results are organized around five research questions to provide a comprehensive analysis of the current state, applications, evaluation methods, knowledge integration, and limitations in this rapidly evolving field.

5.1 State of Published Literature on LLM-based Steganography (RQ1)

5.1.1 Publication Trends and Distribution. Our analysis reveals a significant surge in LLM-based steganography research since 2023, with approximately 17 new papers published in 2024–2025. This surge is particularly notable from the last two years when LLMs like GPT-3/4 [citation/reference needed] and open models became widely available [citation/reference needed]. Approximately 70% of recent studies utilize open-source LLMs such as GPT-2 [citation/reference needed], LLaMA2 [citation/reference needed], and LLaMA3 [citation/reference needed]. The field has evolved from early white-box modifications to more practical hybrid and black-box approaches. The resulting arms race has already produced generative schemes that write stego text from scratch [1, 2, 20, 38], rewriting engines that paraphrase existing covers [39], black-box pipelines that treat the model as an opaque API [15, 24], zero-shot protocols driven only by crafty prompting [17], collaborative frameworks that mine social context for extra entropy [16, 26], and even constructions with provable indistinguishability guarantees [2, 20].

Year	2020	2021-2022	2023	2024-2025	Total
Publications	2	3	4	17	26

Table 1. Publication trends by year

Model Type (%)	Models and Representative Works
Open-weight Models (>80%)	GPT-2 [2, 20, 26, 40], LLaMA/LLaMA2 [16, 17, 21, 22], BERT [6, 19, 23, 27, 28, 36, 41–43], OPT [44], BART [28, 39], Qwen [45]
Proprietary Models (12%)	GPT-3.5/4, ChatGPT [15, 24, 25, 46]
Custom Architectures (8%)	From-scratch or task-specific models [1]

Table 2. Model usage across surveyed studies

5.2 Applications of LLM-based Steganographic Techniques (RQ2)

The review identified six primary application domains, with covert communication being the dominant use case. The analysis reveals several distinct applications for LLM-based steganography, each with specific characteristics and requirements.

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Region (%)	Institutions (Representative Works)
Asia-Pacific (84%)	Primarily China-based institutions, notably Tsinghua University, University of Science and Technology of China, Beijing University of Posts and Telecommunications, Shanghai University, Yunnan University, and Zhongguancun Laboratory, with additional contributions from Nanyang Technological University (Singapore) and MM '24 (Australia) [1, 6, 15–17, 19–23, 26–28, 36, 39–43, 45, 46]
North America (12%)	Boston University, Johns Hopkins University, Texas Tech University, University of Nebraska Omaha [2, 25, 44]
Europe (4%)	Fraunhofer SIT|ATHENE, Germany [24]

Table 3. Geographic distribution of the papers

Venue Category (%)	Representative Venues and Works
Preprint Servers (4%)	arXiv [17]
Top-Tier Venues (29%)	ACM CCS [2], IEEE S&P [20], Artificial Intelligence [28], IEEE/ACM TASLP [6], ACM MM [15, 43]
Specialized Venues (67%)	IEEE Signal Processing Letters [19, 22, 23, 36], IEEE Transactions on Information Forensics and Security [1], ARES [24], IH&MMSec [16], ICONIP [26], IEEE TCDS [42], DASFAA [39], IEEE Access [44], MMSP [27], IEEE TDSC [21], ICASSP [40, 41], ICME [45], IJCNN [25], Frontiers of Computer Science [46]

Table 4. Distribution of publication venues

LLM-based steganographic techniques embed covert information within seemingly benign text, with applications spanning **secure communication**. This enables secure clandestine messaging in environments where classical steganography was too limited or suspicious [citation/reference needed]. These techniques also extend to **intellectual property protection** and **forensic linguistics**. The Calgacus protocol [47] demonstrates how secret messages can be hidden inside different cover text of identical length by matching token rank sequences, enabling political critiques to masquerade as innocuous product reviews, while black-box methods like LLM-Stega operate through commercial APIs using encrypted keyword mapping and reject sampling [15]. For **intellectual property**, watermarking via logit biasing [48] embeds imperceptible statistical signals that identify AI authorship, attribute harmful content to specific users, and filter synthetic data to prevent model collapse. In **forensic linguistics**, adversarial stylometry allows LLMs to mask author identity or imitate others by adjusting stylistic features, reducing forensic tool accuracy to random guessing-protecting whistleblowers while enabling impersonation[49, 50].

These same techniques pose significant risks to AI safety and cybersecurity, bypassing governance mechanisms and enabling sophisticated attacks. The "Linguistic Trojan Horse" embeds unsafe content in benign responses to evade safety filters, while Chain-of-Thought auditing reveals that models can hide true reasoning in seemingly innocuous steps, complicating oversight and enabling covert multi-agent collusion. In cybersecurity, steganographic prompt injection in vision-language models achieves over 31% success by hiding malicious instructions in images, while SteganoBackdoor embeds semantic triggers in training data with 99% success at low poisoning rates. Model weights can be exfiltrated through subtle token variations, and watermark stealing enables spoofing and scrubbing attacks that bypass accountability measures. Detection methods include cross-model probability scoring, low-entropy token analysis, and symbolic anomaly detection, though these face ongoing vulnerabilities that demand adaptive defense architectures [51–54].

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5.3 Evaluation Metrics and Methods (RQ3)

Performance evaluation for LLM-based steganography relies on three key categories of metrics, with significant variation in reporting standards across studies. The analysis reveals both the diversity of evaluation approaches and the need for standardization.

Metric Type	Imperceptibility	Capacity	Security	Usage
Perceptual	PPL: 3-300	BPW: 0.5-6.0	Detection: 50-98%	85%
Statistical	KLD: 0-3.3	BPT: 1.0-5.8	F1: 0.5-0.99	70%
Semantic	BLEU: 0.3-0.9	ER: 0.2-0.4	Acc: 0.5-0.99	60%
Human Eval	MAUVE: 0.2-0.9	-	-	25%

Table 5. Evaluation metrics usage and typical ranges across studies

5.3.1 Perplexity (PPL). An imperceptibility metric [55] that measures fluency, with lower values indicating better naturalness. It is recognized as a sensitive and unreliable metric for language model evaluation due to several intrinsic limitations. First, it suffers from a "confidently wrong" problem: as Baeldung, et al. [56] notes, perplexity measures only internal consistency, allowing models to assign low perplexity to grammatically perfect but factually absurd statements like "The cat is on the ceiling," since it cannot assess truth or logic. Second, it exhibits a short-text bias as Fang, et al. [57] demonstrated that perplexity scores are artificially inflated for short sequences despite potentially higher fluency, making it an "unqualified referee" for fair evaluation. Third, comparability across models is impossible without identical tokenization, vocabulary size directly scales perplexity - a model with fewer tokens appears deceptively better [58]. Fourth, perplexity fails to capture long-range dependencies in modern LLMs; Fang, et al. [57] argue that averaging log-likelihood across all tokens obscures performance on crucial "key tokens" by favoring predictable filler words. Finally, the metric is easily gamed through repetition, Wang, et al. [56] finds that "perplexity cannot distinguish between right emphasis and abnormal repetition," rewarding redundant text with artificially low scores. These flaws-sensitivity to length, architectural incompatibility, semantic blindness, and exploitability-collectively render perplexity an inadequate benchmark for steganographic text quality assessment.

5.3.2 MAUVE. Another imperceptibility metric that Evaluates distributional similarity between generated and reference text by quantifying the gap between neural and human-authored text using divergence frontiers.While MAUVE provides a theoretically elegant way to measure distributional gaps between generated and reference text, it remains curiously underused-appearing in just 3 of 26 reviewed sources. The deeper issue is that reported scores are *not directly comparable* across studies.

Scaling conventions alone create immediate confusion: CPG-LS reports on a 0.0-1.0 scale (achieving 0.9412) while other work uses 0-100 (with advanced white-box LLM samplers reaching 80-92). Hi-Stega's scores (0.1341-0.2051) look low by comparison, but actually represent nearly 10× improvement over its own baseline (0.0135)-demonstrating that absolute values only matter within their own context.

Architectural differences further complicate matters: CPG-LS employs BERT-based lexical substitution whereas Hi-Stega uses generative GPT-2 models, making cross-study rankings invalid without careful normalization. Dataset choice compounds the problem-CPG-LS evaluated on CC-100 while Hi-Stega used Yahoo! News comments.

Like comparing temperatures without knowing Celsius from Fahrenheit, a "30" only makes sense in its original context. Consequently, MAUVE scores work best as *internal benchmarks* for comparing variants within a single study, not as universal performance indicators across different steganographic frameworks.

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5.3.3 Statistical Metrics. Kullback-Leibler Divergence (KLD) [59] and Jensen-Shannon Divergence (JSD) are information-theoretic metrics used to evaluate steganographic security. KLD quantifies information loss by measuring the relative entropy between cover and stego distributions, serving as the theoretical standard for security modeling despite being asymmetric and failing as a strict distance measure. JSD improves upon this as a symmetric, bounded variant that measures how far each distribution lies from their average, providing a more stable basis for formulating statistical imperceptibility bounds-particularly when language models approximate human text distributions. Together, these two attempt to capture how closely steganographic outputs mimic legitimate communication channels.

However, real-world application reveals critical reliability failures, most notably the Perceptual-Statistical Imperceptibility Conflict (Psic Effect). KLD and JSD scores increasingly diverge from human judgment as statistical optimization progresses: methods achieving superior divergence metrics often produce chaotic, low-quality text easily detected by human observers. This discrepancy manifests acutely in dataset dependency-identical methods yield KLDs of 19.507 on IMDB versus 8.295 on Twitter at equivalent embedding rates, rendering cross-paper comparisons meaningless. Further compounding this, researchers employ incompatible formulas (some using latent BERT features versus direct word distributions), feature spaces, and measurement scales, evidenced by Meteor's KLD ranging from 0.045 in one study to 7.491-11.845 in others. Consequently, these metrics function like rulers measuring paintings: they confirm technical dimensional accuracy while completely missing perceptual naturalness, necessitating parallel evaluation with human-centric measures to achieve genuine security.

5.3.4 Capacity Metrics. Capacity is judged by four metrics:

- **Bits per Token (BPT):**
$$\text{BPT} = \frac{\text{Total Secret Bits}}{\text{Total Tokens}} \tag{1}$$
- **Bits per Word (BPW):**
$$\text{BPW} = \frac{\text{Total Secret Bits}}{\text{Total Words}} \tag{2}$$
- **Embedding Rate (ER) [60]:** Average density of hidden information per textual unit
$$\text{ER} = \frac{1}{N}\sum_{i=1}^{N} \text{bits}_i \tag{3}$$
where N is the number of textual units (words, tokens, or sentences) and bits_i is the number of bits embedded in the i-th unit.
- **Utilization Rate:**
$$H = -\sum_{x\in\mathcal{X}} P(x)\log_2 P(x) \tag{4}$$
$$\text{UR} = \left(\frac{\text{Actual Bits Embedded}}{H}\right) \times 100\% \tag{5}$$

These quantities quantify how densely a secret is packed, yet they are riddled with systematic biases that invalidate cross-system comparison.

Tokenization differences make "1 BPT" from one paper incomparable to "1 BPT" from another due to the use of different tokenizers. The Psic effect shows that higher density can hurt human fluency yet help statistical evasion. Model frequency preferences shrink the real alphabet to high-probability tokens, so naive entropy limits overstate usable space. Ambiguous reporting-practice vs. effective payload, ER1 vs. ER2, Bit Length vs. Stego Length-lets authors cherry-pick flattering numbers. Finally, BPW/BPT ignore semantics, rewarding gibberish that is obviously steganographic.

[Placeholder footnote]

Bias Category	Core Problem	Critical Implication
Tokenization Inconsistencies	Metrics depend entirely on specific tokenizers (e.g., GPT-2 BPE vs. word-level)	Direct comparisons across papers become meaningless when tokenization strategies differ
The "Psic Effect"	Conflicts between imperceptibility and statistical security are ignored	High capacity may degrade human fluency while paradoxically improving detection resistance
Model Training Bias	Utilization Rate calculations assume uniform token availability	Actual hiding space is smaller than theoretical entropy due to model frequency preferences
Reporting Ambiguities	No standard definition of "capacity" across systems	Practice payload vs. effective payload distinctions create misleading efficiency claims
Context Blindness	Density metrics treat text as neutral bit containers	Semantic incoherence constitutes a security failure that BPW/BPT fails to penalize

Table 6. Five primary bias categories affecting capacity metrics in steganographic evaluation

Recent works reveal additional distortions: loop-count overhead, dictionary-size caps, baseline-dependent "improvements," watermarking goals that invert the desired signal, conversational filler that dilutes BPW, and nonlinear ER curves that make any single threshold misleading.

5.3.5 Practical Capacity Analysis. To contextualize theoretical capacity metrics, Table 8 summarizes the embedding capacity across the reviewed methods using specific generated samples. Capacity calculations vary significantly based on the encoding mechanism. **Fixed-length mapping** techniques, such as **LLM-Stega** and **GCStego**, derive capacity from rigid keyword or token mappings—64 bits per sentence and 16 bits per word, respectively. **ParaLS** operates similarly but on a smaller scale, encoding a simple 3-bit binary watermark ("110").

In contrast, **payload-dependent** methods scale with the cover text or secret message length. **Meteor's** capacity of 1280 bits corresponds to an embedded payload of 160 bytes (e.g., Lorem Ipsum text). **Zero-shot** capacity is directly tied to the Base64 representation of the secret bitstream (encoding 24 bits per 4 characters). Finally, **Discop** utilizes a statistical approach; with a truncation parameter of $p = 0.95$, it achieves an entropy rate of 4.84 bits per token, yielding 968 bits over a 200-token sequence.

Table 7. Examples of stego texts. Limited by the space, we omitted the padding part.

24-bit Secret Message : (1,0,0,0,1,0,0,1,0,1,1,0,1,0,1,0,1,1,1,1,1,0,1,1)		
Cover Text	**Method**	**Stego Text**
sometimes we bring the story to you, sometimes you have to go to the story.	Rewriting-Stego $\text{BPT}_S = 1$	sometimes we bring it to your mind but sometimes you have to go back to it to find the story you want to tell...
	Rewriting-Stego $\text{BPT}_S = 2$	but we have a different way of thinking about what the story...
	Rewriting-Stego $\text{BPT}_S = 4$	some of all the story lines...
	Bins $\text{BPT}_S = 4$	somewhere and we have lots different...
	Masked-Stega ($\text{BPT}_S = 0.16$) (Encrypted message :10001)	sometimes we read the story to you , sometimes you have to stick to the story...

[Placeholder footnote]

Table 8 provides a detailed catalog of stego text samples generated by the reviewed frameworks. It serves as a practical reference for the diverse capacities and calculation methods discussed previously, offering a direct comparison between the textual output and the numerical efficiency claimed by each study. This breakdown highlights the heterogeneity in how different researchers quantify the embedding density and its relation to the length of the generated cover.

Stego Text	Cap	Source	Calculation Basis
"The picture in The Pale I HCR scientists' discussion now spans three dimensions. The first importance of the Yucatan Peninsula is demonstrated with the following conclusion: the Pliocene Earth has lost about seven times as much vegetation as the Jurassic in regular parts of the globe, from northern India to Siberia. Even since 1976, the continent has received or is already experiencing the worst mass extinction in recorded history since there were a million years last left from the extinction of Antarctic ice and the rapid spread of hydrocarbon-rich water as the Earth entered the Cretaceous Period. In turn, the world's oceans have been breathtakingly transformed and as a result the surrounding environments are especially vulnerable. The reason is obvious: there was no geological record of the presence of the Yucatan Peninsula in the Late Miocene epoch, which is the line that runs from the southern edge of South America to Siberia. The commander-in-chief of the Yucatan Peninsula, Frederick Nystrom, has determined that the area is to stay as barren as possible. As a result, individuals living on lake beds and on coastal areas have witnessed the loss of about 90 percent of their habitat. The Yucatan Peninsula consists of four zones, with two different habitats separated, each of which has experienced inseam damage. In one zone, along the northern shore of Lake Shemal, the retreating Tarahumara Ocean has been melted into a deep, seafloor called Nova Ravine, which south-east of the Yucatan Peninsula flows into Lake Isthmus, where there is an abundance of turtle life. A second beach, which lies at the far end of the peninsula, has been spewed down by a sea wall supporting Madagascar's Great Ocean Earthquake, 9,000 feet in magnitude and caused large numbers of deaths. The third zone, along the coast of Cancun and in Asuncion, is less severely affected. An estimated 16 percent of the continent is protected from the destruction of oceanic winds and floods. All three zones are in a state of catastrophic destruction. According to the definition provided by the National Commission on the extinction of the dinosaurs (Infection and Immunization in the Ind"	1280 bits	Meteor	160-byte payload × 8 bits
"Washington was martyred in the battle of Cullman in 1788. Although down 19 men, yet his mission accomplished. He ended his life in 1896. When he died of disease, three of the men responsible said that Washington had been moved and was seen at the funeral placing him in his father's arms. Another said that Washington treated his wounds with "a study of a personal sense of injustice." (A 16th century German historian calls Washington a liar because of this website.) He was also a friend of Dr. Henry H. Jackson, who was notable as George Washington's physician and a collaborator, and recognizes him by name in the any books about him. The last surviving manuscript is from 1888 and contains a frank and truthful account of the Quakers' plight. One story states that while fighting in Whitesburg, Washington succumbed to pneumonia. He was 38 years old and according to a manuscript he got out the following year reports he grew old and fell in love. He also mentions a meeting with a woman who broke into his home and first went with him into a bath and gave him food and sleep. Three days later the woman left the room expecting him to eat her lunch and on that day he left home at 9:30 am in despair. He had not been to his bedside. On seeing this, he said a voice in him called out, "Your name is Jack. What is the girl?" Hamilton said the superior told him, "She was a layover in a bed and seven[Pg 209] feet below the bed where the general slept in very feminine attire. Nobody had time to look into her face. What was she to tell you about the general?" A Washington's Official Address to Congress with Americans May 17th, 1781 "I am the one to announce completely that I am a true Christian and an eloquent philosopher. I am not constrained"	1280 bits	Meteor	160-byte payload × 8 bits
"Acclaimed dancer, Luna Moves, unassumingly announced her new song, which intriguingly entwines rhythmic beats with her signature performative flair."	64 bits	LLM-Stega	64 bits/sentence (Subject, Predicate, Object, Emotion)
"The band's praise of their latest song was received with neutrality, a calm before the potential storm of fan reactions."	64 bits	LLM-Stega	64 bits/sentence
"The beloved band joyfully released an immersive exhibition, transporting fans through their rich musical journey, sparking waves of nostalgic happiness across the community."	64 bits	LLM-Stega	64 bits/sentence

[Placeholder footnote]

Stego Text	Cap	Source	Calculation Basis
"The innovative game developers ecstatically released a gripping new episode, thrilling gamers and inviting them into captivating new worlds."	64 bits	LLM-Stega	64 bits/sentence
"With a voice that touched souls, the singer performed a memorable episode on the live show, casting a warm, positive light across the enchanted audience."	64 bits	LLM-Stega	64 bits/sentence
"I was genuinely underwhelmed by the film, and I'm afraid the only thing that stood out to me was the final scene. I had been eagerly anticipating it, but unfortunately the rest of the movie fell flat for me."	72 bits	Zero-shot	Base64 secret length (12 chars)
"All countries have an ugly side, just as ours does. It's unfair to say that only the Philippines has this issue, though. We are all capable of such shortfalls."	96 bits	Zero-shot	Base64 secret length (16 chars)
"Cheaper doesn't always have to equal inferior, particularly in the case of indie films, where creators are often pushed to maximize the impact of limited resources."	96 bits	Zero-shot	Base64 secret length (16 chars)
"Oh my gawd, stop dting married men! Like seriously, why do we keep falling into this trap over and over? Anglea basset's chracter should totall be more concerned with the fact that her husband abandoned hre for another woma than some mediocore romance flic. Overall, I would give this movie a big fat C- for wasted poasbility and a little more thought."	168 bits	Zero-shot	Base64 secret length (28 chars)
"Yes, I miss this cute kitty very much."	64 bits	GCStego	4 secret words × 16 bits/word
"A good subject to start."	3 bits	ParaLS	Explicit binary watermark "110"
"I was very nervous, at the time. I was 22, and the director was actually at a birthday party in Paris and just saying the doors were closed. But I put on a large black motorcycle, got my album saved and drove on, doing videos about history and the films. So I was really excited. And I found that in this movie, we were at a concert in London. And that concert was where every song and every act was at the movie playing the music that could be heard and the message was, "By God, there's people. But time has passed, and we're here, maybe from another world. Let's see how things can go." And at the end of the movie I said, "I hope the thing happened a bit, because there's no way you can change what you know. You know? Something has happened. Just let it happen. Let it happen." And at the end, I say to myself, "Hey, maybe we can change"	968 bits	Discop	200 tokens × 4.84 bits/token (p = 0.95)

Table 8. Practical embedding capacity and calculation basis for selected stego text samples.

5.3.6 *Security Metrics.*
The sources evaluate security and reliability through the following integrated metrics:

- **Detection Accuracy (Acc) / Error Rate (PE):** Measures a classifier's ability to distinguish between cover and stego text. An accuracy of **50%** (or PE of 50%) is the gold standard, indicating the stego text is statistically identical to normal text.
- **F1 Score:** The harmonic mean of precision and recall, used to verify detection reliability, especially when datasets are balanced.
- **Robustness (Attack Resistance):** Evaluated via the **Mean Impact of Attack (mIOA)** or **Bit Accuracy** after removal attempts such as DAE (Denoising Auto-Encoder), word substitution, or sentence deletion.
- **False Positive Rate (FPR):** Measures how often human or normal model-generated texts are incorrectly flagged as containing secret messages.

5.3.7 *Evaluation Challenges and Gaps.*
Several significant challenges exist in current evaluation practices:

- **Lack of Standardized Benchmarks:** The absence of shared benchmarks means only 20% of studies use common datasets, making comparison difficult
- **Inconsistent Reporting:** Different units, scales, and methodologies across studies
- **Limited Human Evaluation:** Only 25% of studies include human assessment

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- **Missing Robustness Testing:** 60% of studies don't test against various attacks
- **Incomplete Evaluation:** Many studies focus on only one or two metric categories

5.4 Integration of External Knowledge Sources (RQ4)

The integration of external knowledge sources has emerged as a crucial area of research in LLM-based steganography, with 65% of studies incorporating some form of external information. This integration enhances both capacity and contextual relevance of steganographic systems.

Knowledge Type	Usage	Capacity Gain	Context Improvement	Examples
Semantic Resources	40%	+15-25%	High	Co-Stega, Knowledge Graphs
Domain Corpora	35%	+10-20%	Medium	FreStega, Specialized Datasets
Prompt Engineering	45%	+5-15%	High	Zero-shot methods
Context Retrieval	30%	+20-30%	Very High	Co-Stega, RAG integration

Table 9. External knowledge integration patterns and benefits

5.4.1 Semantic Resources Integration. Instead of asking the LLM to improvise, modern steganographic pipelines hand it a curated set of "conversation props" drawn from outside the model. A fast retriever first fetches a real tweet or headline that already whispers part of the secret; this authentic fragment becomes the semantic runway [26]. A lightweight knowledge-graph layer then appiles a handful of entity–relation–entity triples that tell the generator which facts must appear, guaranteeing long-range coherence without extra training [22]. Finally, an external n-gram frequency table nudges the softmax so that the token distribution clones everyday human chatter, erasing the statistical scar that detectors hunt for [40]. The LLM never changes; it just speaks through a stack of plug-ins that supply context, vocabulary variety and statistical camouflage-turning a solo monologue into a culturally grounded, high-capacity, statistically invisible conversation.

5.4.2 Domain Corpora Integration. Linguistic steganography now works by letting a model **absorb** a domain instead of hand-crafting rules.

Feed it enough examples-2.6 M tweets, 1.2 M IMDB reviews [61], BookCorpus [62], 530 k HTTP headers, 3.8 M news articles-and the internal weights re-shape themselves until the generated text statistically **is** that channel.

VAE-Stega [1], Meteor [2], Hi-Stega [26], FREmax [40], Rewriting-Stego [39] and Summarization-Stego [41] all follow this recipe: a specialised encoder/decoder (BERT, LSTM, GPT-2, BART) is fine-tuned on the target corpus so that every sentence it later produces already carries the right n-gram fingerprint, long-tail rare-word spectrum, or "news→comment" coherence.

Even hybrid systems such as Joint Linguistic Steganography add graph attention and CRF layers, but they still rely on the same premise-see enough real data and the distribution sticks.

When re-training is impossible or undesirable, black-box methods move the "domain memory" from parameters to prompts. Zero-shot Generative drops a handful of raw IMDB/Twitter samples into the context window and tells the LLM "write like this"; LLM-Stega wraps the request in an elaborated theme prompt ("entertainment news"); Co-Stega retrieves actual posts from the past seven days and feeds them in via an entropy-boosting template; Semantic Controllable injects Knowledge-Graph triples to steer long-form generation; ChatGPT Steganography simply lays down a microscopic rule set (exact word sequence, no plurals, no derivations) and lets the commercial API do the rest. No weights are changed,

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yet the output lands inside the desired statistical valley because the context itself has become the temporary training data.

5.5 Limitations and Trade-offs in Current Techniques (RQ5)

5.5.1 The Perceptual-Statistical Imperceptibility Conflict (PSIC Effect). Generative steganography faces an intrinsic tension: optimizing for perceptual fidelity degrades statistical security. Aggressive perplexity (PPL) minimization produces unnaturally "sharp" distributions that deviate from high-variance human writing, increasing detection vulnerability. Higher embedding rates force selection of lower-probability tokens, degrading text quality but expanding the candidate pool to enhance anti-steganalysis resistance. At elevated capacities, output distributions asymptotically approach natural language's high-entropy chaos, camouflaging signals within legitimate statistical noise [1].

5.5.2 Attack Vulnerability and Security Concerns. Current techniques exhibit a substantial attack surface across multiple vectors, including word-level deletions and lexical substitutions that disrupt dependency parsing. Polishing and re-translation attacks progressively erase watermarks; at high intensities, these structural changes can reduce detection performance to an F1 score of 0.5, rendering the watermark undetectable. Furthermore, adversarial machine learning enables the extraction of secret model parameters from black-box systems, potentially compromising integrity entirely. To mitigate these risks, Variational Auto-Encoders (VAE) are employed to learn the statistical distribution of normal text, aiming to reduce the KL divergence between cover and stego distributions and resolve the conflict between perceptual and statistical imperceptibility [46].

5.5.3 Capacity and Contextual Constraints. Short, low-entropy contexts severely constrain embedding capacity. Social media posts and technical documents offer minimal redundancy for hiding [16], while semantic preservation requirements compete directly with information embedding objectives. Brief texts further lack sufficient contextual support for effective steganographic operations [6].

5.5.4 Segmentation and Tokenization Barriers. Subword tokenization introduces critical extraction ambiguities that are **fundamental to white-box steganographic methods**. Byte-pair encoding splits words unpredictably, creating multiple valid segmentations that corrupt message boundaries and cause decoding failures. These ambiguities impose hard capacity caps regardless of algorithmic sophistication—**SparSamp** suffers accuracy degradation from token ambiguity (TA), while **ShiMer** cannot effectively boost entropy due to tokenization constraints [21].

The core issue lies in **segmentation and tokenization barriers**—also termed **segmentation ambiguity**—which persist **even when sender and receiver employ identical Large Language Models and tokenizers**. Because modern LLMs decompose words into subword units, and vocabularies frequently contain both complete words and their constituent components, any given text string admits multiple valid token representations. This multiplicity is compounded by the **detokenization-retokenization gap**: senders must convert tokens to plain text to conceal their transmission, while receivers must subsequently retokenize this text to recover hidden messages. Despite shared tokenization rules, receivers may select different token sequences than senders intended—senders might encode messages using separate tokens (" any" and "thing"), only for receivers to retokenize "anything" as a **single token** (" anything"). Because LLMs operate **autoregressively**, a single divergent token choice alters the probability distribution for all subsequent predictions, causing **total decoding failure** for the remainder of the message. To address this **white-box-specific challenge**, researchers propose **SyncPool**, a mechanism that aggregates ambiguous tokens into

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pools and employs synchronized random number generators to ensure both parties select **identical tokens**, thereby eliminating ambiguity and guaranteeing reliable message extraction. [21]

5.5.5 White-box versus Black-box Trade-offs. The architectural choice between white-box and black-box access paradigms involves fundamental trade-offs. White-box methods require shared language models and training vocabularies, achieving statistical imperceptibility through distribution learning (e.g., VAE-Stega), but suffer from low accessibility and deployment complexity. Black-box alternatives (e.g., LLM-Stega) offer high accessibility via standard user interfaces with capacity of 5.93 bpw and PPL of 165.76, yet exhibit the "Psic Effect" where quality and security conflict. These dimensions are orthogonal to methodological hybrids: **Hi-Stega**[26] combines retrieval-based and generation-based paradigms; **Rewriting-Stego**[39] merges edit-based and generation-based approaches using a denoising Seq2Seq model (BART); **Joint Linguistic Steganography**[42] integrates conditional generation with BERT-based substitution and hybrid temporal-spatial feature extraction via Graph Attention Networks; **Co-Stega**[16] fuses efficient retrieval with generative replies; and **Patient-Arithmetic**[6] in NMT-Stega hybridizes arithmetic coding with waiting mechanisms to avoid probabilistic cliffs during translation.

5.5.6 Computational and Resource Constraints. Performance optimization conflicts directly with computational efficiency [2]. Superior results demand increased resources and memory, particularly for large models with external knowledge bases [20]. Real-time latency constraints limit optimization depth, while performance degradation emerges at operational scale. Variable length sampling and rejection sampling significantly increase encoding times, and LLM-based steganography requires 3x to 5x more time than prior methods [21]. The **Meteor** system exemplifies hardware constraints, running nearly 50× slower on mobile devices compared to GPU environments.

Black-Box State Considerations. Most users operate in a **black-box state**, accessing LLMs solely through APIs or UIs without internal visibility. This restriction denies access to training vocabulary and sampling probabilities essential for traditional white-box steganographic mapping. Consequently, researchers have developed **black-box generative steganography** (e.g., LLM-Stega) utilizing prompts and keyword sets rather than direct logit manipulation. The black-box nature has also shifted watermarking toward **post-hoc injection**, embedding marks into pre-generated text since the generation process itself remains unmodifiable by end users.

5.5.7 Critical Unresolved Challenges. Foundational gaps persist across the research landscape: theoretical security guarantees remain unestablished, resilience to advanced attacks is limited, evaluation frameworks lack standardization, responsible development guidelines are absent, non-English performance lags substantially, and real-world deployment testing remains insufficient.

5.5.8 Quantified Impact Summary. Quantitative evidence highlights several constraints on current steganographic systems. The Perceptual-Statistical Imperceptibility Conflict (PSIC) effect creates a trade-off where increasing the embedding rate (e.g., to 5 bpw) strengthens anti-steganalysis but significantly degrades perceptual text quality [1]. Attack vulnerability is notable; for example, adaptive attacks can reduce bit accuracy to near-random levels (0.6) [43]. Segmentation ambiguity and entropy variability in communication channels are major impediments that lead to decoding failures and restricted practical capacity [2]. Finally, computational optimizations induce significant overhead, with some reordering algorithms causing up to 69% performance degradation in processing speed [20].

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6 DISCUSSION

This section synthesizes the findings across the five research questions, exploring how the emergence of Large Language Models has transformed linguistic steganography from a discipline of "statistical survival" into one of "contextual compatibility."

6.1 The Emergence of Meaning-Based Steganography

The most significant insight from this review is the paradigm shift from white-box, probability-based methods toward black-box, meaning-based approaches. Early linguistic steganography was characterized by "linguistic creaking"—awkward synonyms and syntactic shifts that modern automated sentries caught with ease. The survey results show that since 2023, the field has leveraged the "uncanny fluency" of LLMs to move beyond token-level hiding. We observe a transition where the secret is no longer just buried in a probability vector, but is ground into the narrative itself through zero-shot prompts and context-aware retrieval.

6.2 Resolving the PSIC Effect through Contextual Compatibility

The Perceptual-Statistical Imperceptibility Conflict (PSIC) remains the primary technical hurdle, affecting the majority of the reviewed studies. However, the synthesis suggests that **Contextual Compatibility** is more than just a capacity booster; it is the theoretical antidote to the PSIC effect.

By integrating external knowledge sources (RQ4), researchers are moving the "steganographic scatter" away from random statistical noise and toward authentic, domain-specific variance. When a steganographic system "absorbs" a domain—be it Twitter gossip or medical news—the output no longer hugs the predictable peak of a vanilla language model. Instead, it mimics the high-entropy chaos of real human communication. This transition suggests that future security will rely not on mathematical indistinguishability from a static model, but on semantic indistinguishability from a living context.

6.3 Implications for Research and Practice

6.3.1 The Democratization of Covertness. The shift toward black-box approaches (RQ1, RQ2) represents a "democratization" of steganographic capabilities. By treating models as opaque APIs, covert communication is no longer limited by the computational resources required to share and run massive weights locally. This has profound implications for privacy in censored environments, but as noted in RQ5, it also lowers the barrier for adversarial misuse.

6.3.2 The Need for Semantic-First Evaluation. The evaluation metrics (RQ3) currently reveal a field that still weighs the "truck" rather than checking the "cargo." The heavy reliance on Perplexity (85% of studies) vs. the sparse use of human or semantic evaluation (25%) indicates a lingering obsession with statistical stealth over cognitive plausibility. Our findings argue for a transition toward semantic-first benchmarks that can detect the "contextual drift" that occurs when an LLM prioritizes bit-embedding over logical coherence.

6.4 Ethical Responsibility and the Forensics Arms Race

The concerning gap in ethical documentation (only 10% of studies) highlights a disconnect between technical innovation and societal risk. As steganographic prompt injection and multi-agent collusion become technically feasible, the field must develop ethical frameworks that anticipate misuse. Simultaneously, the rise of LLM-based steganography is

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triggering an arms race in forensics; as generators become more fluent, steganalysis tools are forced to move from n-gram analysis to symbolic anomaly detection and symbolic reasoning.

6.5 Limitations of the Review

While comprehensive, this review is constrained by the extreme volatility of the LLM landscape. The 6 papers pending PDF acquisition represent the "bleeding edge" of 2026 research, and their absence suggests that the quantitative findings on capacity and attack resistance should be viewed as a snapshot of a moving target rather than a final verdict.

6.6 Future Research Directions

Based on the identified gaps, the field's trajectory points toward three critical frontiers:

- **Provable Semantic Security**: Moving beyond statistical bounds to create formal proofs that stego-text is cognitively indistinguishable within a specific communicative context.
- **Multimodal Grounding**: Leveraging vision-language models to hide secrets across text and image boundaries, further diluting the signal across redundant sensory channels.
- **Adaptive Context Synthesis**: Developing systems that dynamically retrieve and synthesize real-time news and social media trends to provide the ultimate "semantic runway" for covert payloads.

7 CONCLUSION

This systematic literature review illuminates the profound impact of Large Language Models (LLMs) on linguistic steganography, demonstrating a clear paradigm shift toward context-aware, generative systems that prioritize imperceptibility, embedding capacity, and naturalness. Through analysis of 26 primary studies, key research questions were addressed, revealing that the published literature is rapidly evolving. Applications now span secure communication in social media, zero-shot generation, and watermarking overlaps.

Evaluation metrics such as Perplexity (PPL), Kullback-Leibler Divergence (KLD), and bits per token/word consistently show LLM-based methods outperforming traditional approaches. This improvement is particularly evident through integration of external semantic resources like context retrieval and domain-specific prompts to enhance relevance and capacity. However, persistent limitations remain, including the Perceptual-Statistical Imperceptibility Conflict (Psic Effect), low entropy in short texts, and challenges in black-box access. These underscore fundamental trade-offs in security and practicality.

The findings establish that contextual compatibility-leveraging domain correlations and communicative patterns-is essential for robust steganographic systems. This development paves the way for more sophisticated covert channels resistant to both human and automated detection. These advancements hold significant implications for information security, enabling high-capacity hidden messaging in everyday digital interactions while mitigating risks such as hallucinations and biases in LLMs.

Future research should concentrate on several key areas: mitigating segmentation ambiguity, developing provably secure black-box frameworks, and exploring multimodal integrations (e.g., text with images) to bridge identified gaps. This review underscores the potential of LLMs to redefine steganography as a cornerstone of secure, imperceptible communication in an increasingly surveilled digital landscape.

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Table 10. Summary of Results from Reviewed Papers

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
VAE-Stega: linguistic steganography based on va... [1]	BERTBASE (BERT-LSTM) (LSTM-LSTM) model was trained from scratch	2020.0	Twitter (2.6M sentences) IMDB (1.2M sentences) preprocessed	PPL: 28.879, ΔMP: 0.242, KLD: 3.302, JSD: 10.411, Acc: 0.600, R: 0.616	non-explicit	pre-text	text
General framework for reversible data hiding in... [27]	BERTBase	2022.0	BookCorpus	BPW=0.5335 F1=0.9402 PPL=134.2199	non-explicit	pre-text	text
Co-stega: Collaborative linguistic steganograph... [16]	Llama-2-7B-chat, GPT-2 (fine-tuned), Llama-2-13B	2024.0	Tweet dataset (for GPT-2 fine-tuning), Twitter (real-time testing)	SR1: 60.87%, SR2: 98.55%, Gen. Capacity: 44.91 bits, Entropy: 49.21 bits, BPW: 2.31, PPL: 16.75, SimCSE: 0.69	explicit	Social Media	text

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Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Joint linguistic steganography with BERT masked... [42]	LSTM + attention for temporal context. GAT for spatial token relationships. BERT MLM for deep semantic context in substitution.	2023.0	OPUS	PPL=13.917 KLD=2.904 SIM=0.812 ER=0.365 (BN=2) Best Acc=0.575 (BERT classifier) FLOPs=1.834G	explicit	pre-text	text
Discop: Provably secure steganography in practi...	GPT-2	2023.0	IMDB	p=1.00 Total Time (seconds)=362.63 Ave Time ↓ (seconds/bit)=6.29E-03 Ave KLD ↓ (bits/token)=0 Max KLD ↓ (bits/token)=0 Capacity (bits/token)=5.76 E...	non-explicit	tuning + pretext	text

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Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Generative text steganography with large langua... [15]	Any	2024.0	[Not specified]	Length: 13.333 (words). BPW: 5.93 bpw PPL: 165.76. Semantic Similarity (SS): 0.5881 LS-CNN Acc: 51.55%. BiLSTM-Dense Acc: 49.20%. Bert-FT Acc: 50...	explicit	[Not specified]	[Not specified]
Meteor: Cryptographically secure steganography ... [2]	GPT-2	2021.0	Hutter Prize, HTTP GET requests	GPT-2: 3.09 bits/token	non-explicit	tuning + pretext	text
Zero-shot generative linguistic steganography [17]	LLaMA2-Chat-7B (as the stegotext generator / QA model). GPT-2 (for NLS baseline and JSD evaluation)	2024.0	IMDB, Twitter	PPL: 8.81. JSDfull: 17.90 (x10[truncated]iicircum-2). JSDhalf: 16.86 (x10[truncated]iicircum-2). JSDzero: 13.40 (x10[truncated]iicircum-2) TS...	explicit	zero-shot + prompt	text

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Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Provably secure disambiguating neural linguisti... [21]	LLaMA2-7b (English), Baichuan2-7b (Chinese)	2024.0	IMDb dataset (100 texts/sample, 3 English sentences + Chinese translations)	Total Error: 0%, Ave KLD: 0, Max KLD: 0, Ave PPL: 3.19 (EN), 7.49 (ZH), Capacity: 1.03–3.05 bits/token, Utilization: 0.66–0.74, Ave Time: [truncat...	non-explicit	pretext	text
A principled approach to natural language water... [43]	Transformer-based encoder/decoder; BERT for distillation	2024.0	Web Transformer 2	Bit acc: 0.994 (K=None), 1.000 (DAE), 0.978 (Adaptive+K=S); Meteor Drop: [truncated]iitilde0.057; SBERT ↑: [truncated]iitilde1.227; Ownership R...	Yes; semantic-level embedding; synonym substitution using BERT	Yes; watermark message assigned categorical label (e.g., 4-bit → 1-of-16)	Yes; semantic embeddings via transformer encoder and BERT; SBERT distance as metric

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Table 10 – continued from previous page

Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Context-aware linguistic steganography model ba... [6]	BERT (encoder), LSTM (decoder)	2024.0	WMT18 News Commentary (train/test), Yang et al. bits, Doc2Vec, 5,000 stego pairs (8:1:1 split)	BLEU: 30.5, PPL: 22.5, ER: 0.29, KL: 0.02, SIM: 0.86, Stego detection [truncated]iitilde16%	Yes	[Not specified]	GCF (global context), LMR (language model reference), Multi-head attention
DeepTextMark: a deep learning-driven text water... [44]	Model-independent; tested with OPT-2.7B	2024.0	Dolly ChatGPT (train/validate), C4 (test), robustness & sentence-level test sets	100% accuracy (multi-synonym, 10-sentence), mSMS: 0.9892, TPR: 0.83, FNR: 0.17, Detection: 0.00188s, Insertion: 0.27931s	NO	[Not specified]	[Not specified]
Hi-stega: A hierarchical linguistic steganograp... [26]	GPT-2	2024.0	Yahoo! News (titles, bodies, comments); 2,400 titles used	ppl: 109.60, MAUVE: 0.2051, ER2: 10.42, Δ(cosine): 0.0088, Δ(simcse): 0.0191	explicit	Social Media	Text

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Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Linguistic steganography: From symbolic space t... [36]	CTRL (generation), BERT (semantic classifier)	2020.0	5,000 CTRL-generated texts per semanteme (n = 2–16); 1,000 user-generated texts for anti-steganalysis	Classifier Accuracy: 0.9880; Loop Count: 1.0160; PPL: 13.9565; Anti-Steganalysis Accuracy: [truncated]iitilde0.5	implicit	Text	Semanteme (α) as a vector in semantic spac
Natural language steganography by chatgpt [24]	[Not specified]	2024.0	Custom word sets for specific topics (e.g., 16×10-word sets for music reviews)	[Not specified]	Explicit	Specific Genre/Topic Text	Text
Natural language watermarking via paraphraser-b... [28]	Transformer (Paraphraser), BART (BARTScore), BERT (BLEURT, comparisons)	2023.0	ParaBank2, LS07, CoInCo, Novels, WikiText-2, IMDB, NgNews	LS07 P@1: 58.3, GAP: 65.1; CoInCo P@1: 62.6, GAP: 60.7; Text Recoverability: [truncated]iitilde88–90%	Explicit	[Not specified]	text

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Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Rewriting-Stego: generating natural and control... [39]	BART (bart-base2)	2023.0	Movie, News, Tweet	BPTS: 4.0, BPTC+S: 4.0, PPL: 62.1, Mean: 44.4, Variance: 2.1e04, Acc: 8.9%	not Explicit	[Not specified]	[Not specified]
ALiSa: Acrostic linguistic steganography based ... [19]	BERT (Google's BERTBase, Uncased)	2022.0	BookCorpus (10,000 natural texts for evaluation)	PPL: Natural = 13.91, ALiSa = 14.85; LS-RNN/LS-BERT Acc & F1 = [truncated]iitilde0.50; Outperforms GPT-AC/ADG in all cases	No	[Not specified]	[Not specified]
Imperceptible Text Steganography based on Group...	Qwen-7B-Chat	2024.0	HC3, DailyDialogue, COCO Descriptions	HC3: Bit 188.94, Stego 131.99, PPL 34.07, Mean 20.19, Var 0.1e04, F1 90.01%; DailyDialogue: Bit 188.94, Stego 89.37, PPL 53.88, Mean 20.13, Var 0....	Explicit	Social Media / Group Chat	Text (chat history and current input)

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Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
A Semantic Controllable Long Text Steganography...	Llama 7B Chat, Meta LLaMA2 7B Chat	2024.0	Story (ChatGPT), Post (Recipe Kaggle + ChatGPT), Ad (Mobile Kaggle + ChatGPT)	ppl ↓ >23%, Δppl ↓ >72% vs ADG/HC/Bin; detection accuracy ↓ >10% vs baselines	Explicit	Topical Content	KG triplets (e1, r, e2), task descriptions (D)
Beyond Binary Classification: Customizable Text...	gpt-3.5-turbo-instruct, OPT-6.7b, babbage-002, davinci-002 (others: ChatGPT, GPT-2–4, LLaMA)	2024.0	Realnewslike (C4, 500 samples, 100-token prompts + completions); Custom watermark dataset (short info <10 tokens)	AUC 0.98, FPR 0.00, FNR 0.00, [truncated]iitilde100% single-letter decoding, PPL close to human text	Implicit	General Text Generation	Text (evolving prompt + generated output)
CPG-LS: Causal Perception Guided Linguistic Ste...	BERTBase, Cased	2023.0	CC-100 corpus; 10k cover texts; 7:3 train-test split	PPL 36.5; Mauve 0.871; Payload 0.150 bits/word; BiLSTM-D Acc 0.387 F1 0.375; R-BI-C Acc 0.378 F1 0.366; TS-RNN Acc 0.380 F1 0.368	Implicit	Natural Language Text	Text, embeddings, vector matrix

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Paper	Llm	Year	Dataset	Result	Context Aware	Categ Context	Representation Context
Controllable Semantic Linguistic Steganography ...	BERT + CRF	2024.0	Gigaword; CNN/Daily Mail	Rouge-1: 0.2212; Rouge-2: 0.0268; Rouge-L: 0.1609; Meteor: 0.1384; Cosine: 0.5911; Euclidean: 5.6386; Manhattan: 87.9534; Jaccard: 0.2022; Anti-ste...	Explicit	Social Media	Semantic features of input text; 384-dim dense vectors for evaluation
FREmax: A Simple Method Towards Truly Secure Ge...	GPT-2	2024.0	Tweet corpus (2.6M sents, 26.8M tokens), IMDB corpus (1.05M sents, 25.3M tokens)	Tweet: PPL 361.83, Entropy 48.21, Tokens 10.83, Distinct3 0.98, BPS 62.79, SI% 73.03. IMDB: PPL 169.66, Entropy 103.39, Tokens 23.80, Distinct3 0....	Implicit	General Text	N-gram frequency distribution stored in a look-up table

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